

CostView 360 - Comprehensive Product Requirements Document (PRD)

Version: 2.0 (Expanded Architecture & Implementation Guide)

Status: Approved for Development

1. Executive Summary & Vision

CostView 360 is a flagship construction cost intelligence platform designed to eliminate the fragmentation between physical site operations and commercial financial tracking. By unifying budget tracking, procurement, material inventory, labour productivity, and project progress into a single Command Center, CostView 360 provides executives, project managers, and quantity surveyors with real-time, actionable insights.

This PRD details the 10X expansion of the platform's capabilities, breaking down the application into high-fidelity functional requirements, complex multi-stakeholder workflows, and a scalable, robust technical architecture leveraging modern web and cloud technologies.

2. Target Personas & Access Matrices

The platform serves multiple distinct user types, each with tailored interfaces and permission bounds.

Persona	Primary Objectives	Key Modules Accessed
Project Manager (PM)	Oversee project timeline, approve variations, ensure budget compliance.	Budget, Progress, Variations, Reports Studio
Quantity Surveyor (QS)	Manage BOQ, assess cost impact of variations, certify subcontractor valuations.	Commercial Mode, Budget, Subcontractors, Final Account
Procurement Officer	Handle supplier enquiries, issue POs, track deliveries, negotiate	Procurement, Materials, Supplier Performance

Persona	Primary Objectives	Key Modules Accessed
	rates.	
Site Engineer / Storekeeper	Log daily progress, manage GRNs, track labour attendance, log snags.	Materials (GRN), Progress (Diary), Labour (Attendance)
Executive / Developer	Monitor project margin, feasibility, financing, and high-level health.	Command Center, Reports, Project Margin, Developer Mode

3. Deep Dive: Site Operations Module

The Site Operations module is the execution engine of the project, tracking the day-to-day physical and financial reality.

3.1 Budget Management

- **Bill of Quantities (BOQ) Integration:** Live parsing of uploaded BOQ data. Support for multi-level hierarchical cost codes.
- **Live Cost Control:** Real-time variance calculation tracking Approved Budget vs. Committed Cost (POs) vs. Actual Cost (Paid).
- **Predictive Forecasting:** Trend analysis using historical burn rates to project Estimate at Completion (EAC) and Estimate to Complete (ETC).
- **Final Account Reconciliations:** Automated aggregation of variations, prime cost sums, and provisional sums into a final closing ledger.

3.2 Procurement Lifecycle

- **Requisitions Workflow:** Auto-trigger requisitions based on material reorder levels.
- **Supplier Enquiry Board:** Side-by-side bid comparisons tracking lead times, payment terms, and historical supplier quality scores.
- **Three-Way Financial Match:** Automated validation between the Purchase Order (PO), Goods Received Note (GRN), and Supplier Invoice. Flagging discrepancies (e.g., 190 bags delivered vs 200 bags invoiced).

- **Supplier Leaderboards:** Automated scoring based on On-Time Delivery (OTD), Quality Pass Rates, and Price Competitiveness.

3.3 Materials & Inventory

- **Live Stock Gauges:** Visual tracking of available, reserved, and consumed stock levels per SKU.
- **Site-to-Site Transfers:** Kanban-style tracking of material movement across multi-location projects.
- **Variance Logging:** Dedicated workflows for logging damaged goods, over-issues, and reconciliation discrepancies.

3.4 Labour & Productivity

- **Attendance Matrix:** Weekly grid inputs for workers, crews, and trades. Integration with biometric site scanners (future-proofing).
- **Automated Payroll Calculation:** Basic pay, overtime multipliers, and automated deduction processing.
- **Crew Productivity Scoring:** Cross-referencing BOQ output metrics against labour hours to calculate real-time efficiency metrics (e.g., blocks laid per man-hour).

3.5 Progress, Quality & Safety

- **Digital Site Diary:** Time-stamped daily logs cross-referenced with local weather API data and workforce headcount.
- **Snagging & NCRs:** Visual Kanban board for Non-Conformance Reports (NCRs), tracking issues from discovery to remediation and sign-off.
- **Safety Scorecards:** Hero-metric displays for "Days Without Lost-Time Incident" and near-miss logs.

4. Deep Dive: Commercial & Developer Mode

This module adapts to the corporate structure of the user, splitting into Contractor and Developer contexts.

4.1 Contractor Commercial Operations

- **Tender & Estimating:** Pre-construction cost build-ups, preliminary pricing, and margin application.
- **Valuation & Certification:** Generation of interim payment applications to the client based on verified physical progress.

- **Claims & Entitlements:** Documenting loss and expense, linking site diary delays (weather, client changes) directly to Extension of Time (EOT) requests.

4.2 Developer / Asset Owner Operations

- **Feasibility & Acquisition:** Land cost tracking, ROI projections, and yield analyses.
- **Project Finance & Capital Stack:** Tracking equity drawdown, development loans, and interest capitalization.
- **Sales & Receivables:** Unit pricing, buyer ledgers, payment schedules, and collection status tracking.

5. Technical Architecture & Backend Flow

The system requires a highly scalable, real-time architecture capable of processing thousands of financial transactions, material movements, and user events securely.

5.1 Core Technology Stack

- **Frontend Web Application:** Built using **Next.js**, **React**, and **TypeScript**. Styled comprehensively with **Tailwind CSS** to replicate the high-fidelity UI specifications.
- **Backend Services & API:** Developed in **Python** (using FastAPI or Django), enabling rapid data processing, complex mathematical cost aggregations, and integration with AI libraries (e.g., TensorFlow) for predictive forecasting.
- **Primary Database (Relational):** Hosted on **Supabase (PostgreSQL)** to ensure ACID compliance, strict relational integrity for financial ledgers, and Row Level Security (RLS).
- **Real-Time Event Sync:** Powered by **Firebase** Realtime Database and Cloud Messaging to push instant updates to the Notification Bell, Snag Kanban boards, and collaborative Site Diaries.

5.2 Database Schema Overview (Supabase/PostgreSQL)

Key entity relationships ensuring data integrity across the platform:

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-- Simplified Core Schema Representation
```

```
Table: projects
```

```
- id (UUID, PK)
```

```
- name (VARCHAR)
```

```
- type (ENUM: 'contractor', 'developer')
```

```
- budget_total (DECIMAL)
```

```
Table: boq_items
```

- id (UUID, PK)
- project_id (FK -> projects.id)
- code (VARCHAR)
- description (TEXT)
- rate (DECIMAL)
- quantity (DECIMAL)
- category (ENUM: 'Material', 'Labour', 'Plant')

Table: purchase_orders

- id (UUID, PK)
- boq_item_id (FK -> boq_items.id)
- supplier_id (FK -> suppliers.id)
- status (ENUM: 'Draft', 'Approved', 'Dispatched', 'Delivered')
- total_value (DECIMAL)

Table: goods_received_notes (GRN)

- id (UUID, PK)
- po_id (FK -> purchase_orders.id)
- quantity_received (DECIMAL)
- discrepancy_flag (BOOLEAN)

5.3 Critical Backend Workflows

Flow 1: The Three-Way Match (Invoice Approval)

1. **Trigger:** Supplier submits an Invoice payload via API or manual entry.
2. **Python Backend Processing:** The API fetches the corresponding Purchase Order and GRN records from Supabase.
3. **Validation Engine:** The system compares:
 - PO.quantity vs GRN.quantity_received vs Invoice.billed_quantity
 - PO.agreed_rate vs Invoice.unit_rate
4. **Outcome Routing:**
 - *If Perfect Match:* Update Invoice status to 'Matched', trigger Webhook to Accounting (e.g., QuickBooks), and push Firebase notification to Finance Officer.
 - *If Discrepancy:* Update status to 'Discrepancy', lock payment release, and push Firebase alert to Procurement Officer outlining the exact variance.

Flow 2: Automated AI Cost Forecasting

1. **Trigger:** Weekly CRON job or manual refresh.

2. **Data Aggregation:** Python backend pulls time-series data of material consumption rates, labour efficiency scores, and current weather delays.
3. **TensorFlow Model Execution:** Feeds historical baseline data through the predictive model to calculate potential cost overruns for ongoing BOQ items.
4. **Insight Generation:** Writes records to the ai_insights table, feeding the Reports Studio and highlighting risks in the Executive Command Center.

6. Integrations & Webhooks

- **Accounting & ERP:** Native two-way sync with QuickBooks and Xero for Accounts Payable (Invoices) and Accounts Receivable (Certifications).
- **Document Storage:** Automated PDF generation for POs and Reports, syncing directly to Google Drive / SharePoint via OAuth2 tokens.
- **Communication:** Email via SendGrid and WhatsApp Business API integrations for urgent supplier notifications and internal approval requests.

7. Non-Functional Requirements

Category	Requirement Specification
Security & Authentication	JWT-based authentication. AES-256 encryption at rest. TLS 1.3 in transit. Full Row-Level Security (RLS) in Supabase.
Scalability & Performance	Support for concurrent multi-tenant architecture. API response time < 200ms for 95th percentile requests. Dashboard loads under 1.5 seconds.
Availability & Compliance	99.99% Uptime SLA. SOC2 Type II compliance. ISO 27001 data handling standards. Daily automated database backups with 30-day retention.

8. Implementation & Rollout Strategy

To mitigate risk and ensure maximum user adoption, the platform will be developed and released in strategic phases.

- **Phase 1: Foundation (Months 1-3):** Supabase schema design, Next.js UI scaffolding, Authentication, Core BOQ import, and basic Budget tracking.
- **Phase 2: Operations & Real-Time (Months 4-6):** Procurement workflows, Firebase real-time notifications, Materials inventory, and Labour logs.
- **Phase 3: Commercial & Integrations (Months 7-8):** Developer/Contractor toggles, QuickBooks integration, and Document generation.
- **Phase 4: Intelligence (Months 9-10):** Python-based predictive analytics, Reports Studio, and advanced Supplier Performance algorithms.